

## LOLLAND FALSTER MARIBO, Denmark

WMO: 061430

Lat: 54.699N

Lon: 11.440E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 86-98

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 15.6       | 20.2 | 10.3                          | 9.3 | 17.4 | 15.2 | 11.9 | 21.2 | 34.4                  | 41.8 | 31.6 | 40.1 | 10.0                  | 90   | 0.695 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 14.0                      | 79.6            | 67.3 | 75.6 | 65.7 | 72.0 | 64.1 | 69.1                | 76.1 | 67.1 | 73.5 | 65.3 | 71.2 | 11.1                    | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.4                            | 97.3 | 72.1 | 64.6 | 91.2 | 70.0 | 62.9 | 85.8 | 68.1 | 33.1          | 76.4 | 31.5 | 73.8 | 30.2 | 71.1 | 75.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 29.1          | 25.7 | 23.3 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                        |         | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|---------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |         | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg   | 46.9        | 33.4  | 34.3  | 37.2  | 43.7  | 51.6  | 57.7  | 61.8  | 62.5  | 55.8  | 47.9  | 40.4  | 35.4  |  |  |
|                                                                        | DBStd   | 11.87       | 7.44  | 6.79  | 5.65  | 5.76  | 4.95  | 5.00  | 4.70  | 5.40  | 4.21  | 5.43  | 5.51  | 6.25  |  |  |
|                                                                        | HDD50   | 2448        | 514   | 441   | 397   | 207   | 38    | 1     | 0     | 0     | 5     | 104   | 289   | 452   |  |  |
|                                                                        | HDD65   | 6685        | 979   | 861   | 861   | 639   | 416   | 227   | 122   | 117   | 276   | 532   | 738   | 917   |  |  |
|                                                                        | CDD50   | 1310        | 0     | 0     | 0     | 17    | 87    | 232   | 367   | 389   | 180   | 37    | 1     | 0     |  |  |
|                                                                        | CDD65   | 74          | 0     | 0     | 0     | 0     | 0     | 7     | 25    | 41    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74   | 443         | 0     | 0     | 0     | 0     | 1     | 46    | 167   | 225   | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80   | 67          | 0     | 0     | 0     | 0     | 0     | 6     | 33    | 28    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg   | 12.4        | 13.5  | 13.9  | 14.3  | 13.1  | 11.9  | 10.9  | 10.9  | 10.8  | 11.7  | 12.4  | 12.4  | 12.8  |  |  |
|                                                                        | PrecAvg | 23.5        | 2.0   | 1.5   | 1.5   | 1.3   | 1.7   | 2.2   | 2.2   | 2.5   | 2.3   | 2.0   | 2.2   | 2.1   |  |  |
| (15) Precipitation                                                     | PrecMax | 28.3        | 3.7   | 3.3   | 3.4   | 3.0   | 3.4   | 4.8   | 5.6   | 7.0   | 5.6   | 4.4   | 5.0   | 4.0   |  |  |
|                                                                        | PrecMin | 15.9        | 0.0   | 0.1   | 0.3   | 0.1   | 0.4   | 0.0   | 0.6   | 0.6   | 0.5   | 0.9   | 0.6   | 0.9   |  |  |
|                                                                        | PrecStd | 3.4         | 0.9   | 0.6   | 0.7   | 0.6   | 0.8   | 1.1   | 1.2   | 1.6   | 1.4   | 0.7   | 1.2   | 0.9   |  |  |
|                                                                        | PrecStd | 3.4         | 0.9   | 0.6   | 0.7   | 0.6   | 0.8   | 1.1   | 1.2   | 1.6   | 1.4   | 0.7   | 1.2   | 0.9   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%    | DB          | 47.7  | 47.7  | 53.4  | 66.3  | 71.9  | 79.0  | 84.6  | 84.6  | 71.8  | 63.0  | 54.4  | 49.7  |  |  |
|                                                                        |         | MCWB        | 46.6  | 46.5  | 49.2  | 56.9  | 62.9  | 67.0  | 70.2  | 68.6  | 65.1  | 60.6  | 52.5  | 48.7  |  |  |
|                                                                        | 2%      | DB          | 44.9  | 45.2  | 49.6  | 60.8  | 68.0  | 74.3  | 79.1  | 80.6  | 66.6  | 59.2  | 51.1  | 46.8  |  |  |
|                                                                        |         | MCWB        | 43.9  | 44.0  | 46.7  | 54.4  | 59.9  | 64.4  | 67.3  | 67.0  | 60.9  | 56.4  | 49.3  | 45.9  |  |  |
|                                                                        | 5%      | DB          | 43.5  | 43.6  | 46.6  | 56.6  | 64.2  | 71.2  | 75.0  | 76.8  | 64.5  | 57.4  | 49.0  | 44.8  |  |  |
|                                                                        |         | MCWB        | 42.6  | 42.5  | 44.7  | 51.8  | 56.6  | 62.7  | 65.3  | 66.3  | 59.7  | 55.0  | 47.9  | 44.2  |  |  |
|                                                                        | 10%     | DB          | 42.2  | 42.5  | 44.6  | 52.6  | 60.6  | 67.7  | 71.4  | 73.1  | 62.4  | 55.5  | 47.0  | 43.2  |  |  |
|                                                                        |         | MCWB        | 41.5  | 41.4  | 43.0  | 48.8  | 55.0  | 61.5  | 63.7  | 65.2  | 58.1  | 53.3  | 46.0  | 42.6  |  |  |
|                                                                        | 0.4%    | WB          | 46.4  | 46.9  | 49.8  | 58.5  | 63.8  | 69.0  | 72.0  | 71.4  | 66.0  | 60.4  | 53.1  | 48.7  |  |  |
|                                                                        |         | MCDB        | 47.0  | 47.6  | 52.0  | 64.6  | 71.4  | 75.1  | 81.3  | 79.0  | 69.9  | 62.3  | 54.7  | 49.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%      | WB          | 44.0  | 44.3  | 47.0  | 54.7  | 60.5  | 65.9  | 69.3  | 69.1  | 62.6  | 57.6  | 50.0  | 46.1  |  |  |
|                                                                        |         | MCDB        | 44.6  | 44.9  | 48.9  | 59.6  | 66.9  | 71.8  | 77.0  | 76.0  | 65.3  | 59.2  | 50.9  | 46.7  |  |  |
|                                                                        | 5%      | WB          | 42.8  | 42.7  | 44.8  | 52.2  | 57.6  | 64.0  | 67.0  | 67.3  | 60.3  | 55.6  | 47.9  | 44.3  |  |  |
|                                                                        |         | MCDB        | 43.3  | 43.3  | 46.2  | 56.2  | 62.6  | 69.2  | 72.8  | 74.4  | 63.1  | 57.0  | 48.7  | 44.7  |  |  |
|                                                                        | 10%     | WB          | 41.3  | 41.2  | 43.0  | 49.5  | 55.4  | 62.0  | 64.9  | 65.6  | 58.9  | 54.0  | 46.1  | 42.7  |  |  |
|                                                                        |         | MCDB        | 41.8  | 42.0  | 44.1  | 52.3  | 59.6  | 66.6  | 70.1  | 72.2  | 61.3  | 55.4  | 46.8  | 43.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB   | MDBR        | 6.0   | 6.1   | 8.5   | 11.4  | 13.4  | 13.0  | 14.0  | 14.0  | 10.3  | 8.9   | 6.4   | 5.8   |  |  |
|                                                                        |         | MCDBR       | 6.3   | 6.3   | 11.5  | 16.6  | 18.5  | 19.1  | 20.6  | 18.7  | 13.0  | 9.5   | 6.9   | 5.1   |  |  |
|                                                                        |         | MCWBR       | 6.0   | 5.8   | 9.3   | 11.1  | 11.6  | 10.7  | 10.5  | 8.3   | 7.9   | 7.5   | 6.7   | 5.1   |  |  |
|                                                                        |         | MCDBR       | 6.1   | 6.2   | 10.5  | 16.0  | 16.7  | 17.1  | 18.2  | 17.4  | 12.0  | 8.3   | 6.8   | 5.0   |  |  |
|                                                                        | 5% WB   | MCWBR       | 5.9   | 5.8   | 8.8   | 10.9  | 11.2  | 10.6  | 10.6  | 8.9   | 8.2   | 7.2   | 6.8   | 5.1   |  |  |
|                                                                        |         | taub        | 0.319 | 0.341 | 0.369 | 0.404 | 0.396 | 0.392 | 0.411 | 0.411 | 0.382 | 0.360 | 0.340 | 0.310 |  |  |
|                                                                        |         | taud        | 2.304 | 2.319 | 2.275 | 2.233 | 2.339 | 2.379 | 2.340 | 2.349 | 2.412 | 2.419 | 2.358 | 2.290 |  |  |
|                                                                        |         | Ebn at Noon | 194   | 233   | 252   | 259   | 269   | 272   | 263   | 255   | 247   | 224   | 186   | 170   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg  | 152         | 348   | 770   | 1314  | 1677  | 1783  | 1662  | 1335  | 962   | 497   | 199   | 109   | 8     |  |  |
|                                                                        | RadStd  | 23          | 58    | 73    | 158   | 180   | 120   | 187   | 93    | 94    | 64    | 26    | 8     |       |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (63 neighbors) | +0.68 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -293  | N/A   | N/A   |

Nomenclature: See separate page